Shared conceptual representations for words and pictures across semantic tasks

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A core goal of cognitive science is to determine the representations that comprise the mind. Here, we describe a series of behavioral experiments that probe the existence of shared conceptual representations across input modalities and semantic tasks for the same concrete objects (Figure 1–Shared). Early psychological work posited that conceptual representations were *lexical* in format; accessing concepts from another modality, like vision, required a transformation to that lexical form (Paivio, 1971; Figure 1–Lexical). Potter and Faulconer (1975) provide evidence against this theory. They first show that reaction times to name pictures, a task that requires an explicit transformation of input to a lexical form, are systematically longer than words. When asked to categorize the same words and pictures, there are no modality differences. This interaction between modality and task along with additional behavioral (e.g., Potter et al., 1986), computational (e.g., Plaut et al., 2002) and neuroimaging evidence (Fedorenko et al., 2024), suggest that language is not the vehicle of conceptual thought.

It is possible that concepts are represented in a modality-specific (Figure 1–Separate) or task-specific (Figure 1–Task Separate) form. To adjudicate between these possibilities, we introduce a behavioral paradigm in which participants respond to two different semantic tasks when presented with inputs in two different modalities (e.g., words and pictures). If representations are shared across both modalities and tasks, then patterns of reaction times between modalities should be similar across tasks. Specifically, there should be *no* interaction between modality and semantic tasks, and participant-wise reaction times should correlate across modalities and tasks.

Experiment #1 served as a positive control and established what patterns of reaction times should be if concepts are shared across modalities and tasks. 113 participants answered “Did this exist 1000 years ago?” and “Does this make a characteristic sound?” for 50 object concepts presented as written or spoken words (Figure 2). Both modalities access the same lexical representation, thus patterns in reaction times to speech and text will be characteristic of shared representations. Using a generalized linear multilevel model (family = inverse gaussian; link = identity) and random intercepts for each subject and item, we identified no significant interaction between modality and task (β = -1.39, p = 0.505). But, lack of significance does not mean lack of an effect. We therefore adopted an equivalence test to identify whether or not the interaction term was meaningful (with a minimum meaningful effect of 5ms). Using a two-one-sided z-test (TOZT), we found the interaction term to be equivalent to zero (z = 1.728, p_{lower} = 0.042; z = -3.060, p_{upper} = 0.001).

In Experiment #2, the critical experiment, 292 participants answered the same two conceptual questions, but for words and their image counterparts. If concepts are shared across modalities and tasks, we should observe no interaction between the two, as established by Experiment #1. We indeed identified no significant interaction between modality and task (β = -0.48, p = 0.678) (Figure 2) and validate this with the TOZT (z = 3.924, p_{lower} < 0.001; z = -4.754, p_{upper} < 0.001). These results support our hypothesis that concepts are shared across input modalities and conceptual tasks.

In addition, we calculated the correlation between modalities and tasks for each experiment. If concepts are shared across both modalities and tasks within individuals, there should be a positive correlation between them. We find that both speech and text (r = 0.55, p < 0.001) and word and image (r = 0.59, p < 0.001) were positively correlated across participants and task #1 (1000yrs) and task #2 (sound) were positively correlated for Experiment 1 (r = 0.85, p < 0.001) and Experiment 2 (r = 0.84, p < 0.001) across participants (Figure 3).

Altogether we show that reaction times to words and images (1) do not depend on the task, (2) mimic those for speech and text stimuli under the same task constraints, and (3) are positively correlated. We argue that these results support the shared conceptual representation hypothesis (Figure 1–Shared): a single mental representation supports semantic cognition for different tasks and input modalities.

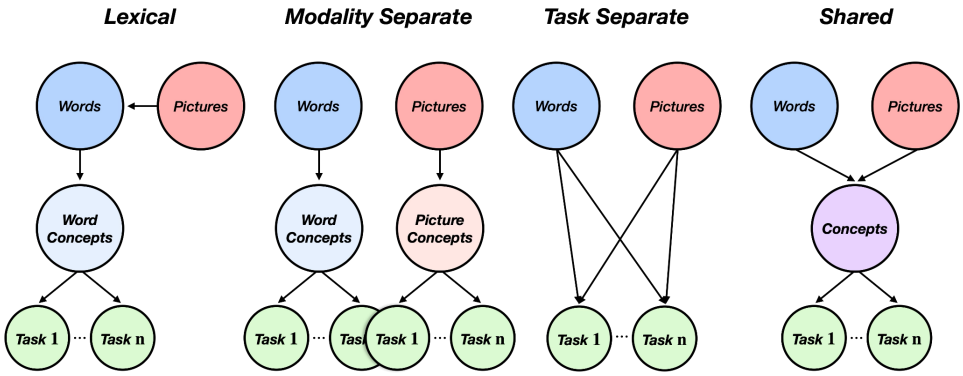

Figure 1 – Possible structures of conceptual representations across modalities and tasks. Concepts may either be lexical in nature (***Lexical***) or are separate across modalities: there is unique concept representation for the image of a dog and for the word "dog" (***Modality Separate***). Perhaps, each task may recruit a task-specific representations (***Task Separate***). Alternatively, words and images may converge on the same underlying conceptual representation across all tasks (***Shared***).

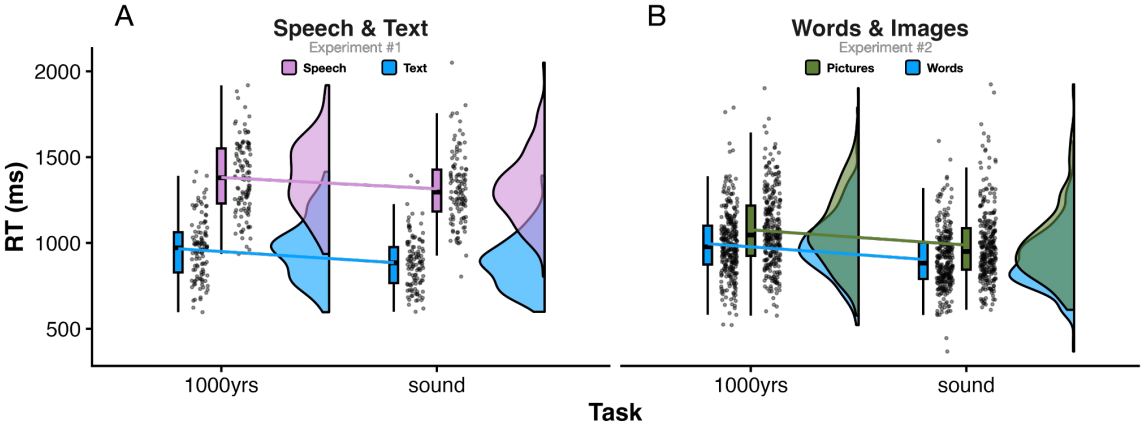

Figure 2 – Reaction times across modalities and tasks. In two separate experiments, participants responded to two conceptual questions (also referred to as "tasks") for 50 items presented in word or image format (***Words & Images***) or speech or text format (***Speech & Text***), the former serving as the main experiment and the latter serving as the control. Individual points are a participant's average reaction time to all items. Solid lines connect mean reaction times for one modality at one task to another, thus visualizing the absence of an interaction.

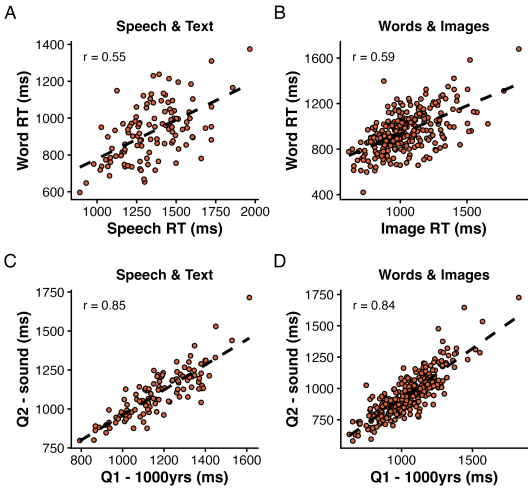

Figure 3 – Correlation of subject reaction times. Correlation of reaction times across words and images (A), speech and text (B) and task 1 and task 2 for Experiment #1 (C) and Experiment #2 (D). Each point corresponds to a single subject.

References

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